

Fuzzy Logic in Practice: A Smart Vent Speed Controller

ESE 561 – Theory of Artificial Intelligence · Problem Solving

The Scenario

A smart HVAC vent adjusts its fan speed based on how far the room temperature has drifted above the thermostat's setpoint. A classical (crisp) controller would only be able to switch the fan OFF, LOW, or HIGH at fixed temperature thresholds — causing abrupt jumps. A **fuzzy logic controller** instead reasons in degrees of membership, producing a smooth, continuously varying fan speed. Figure 1 shows the control loop.

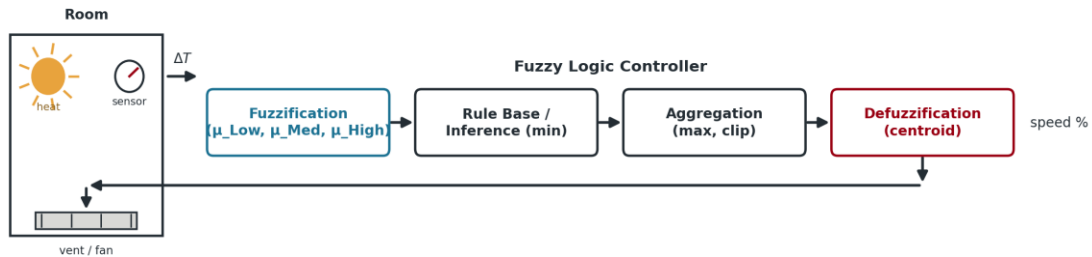


Figure 1. Fuzzy control loop for the vent speed controller.

The Fuzzy Sets

The input variable is the temperature increase ΔT ($^{\circ}\text{C}$ above setpoint), described by three triangular sets — **Low**, **Medium**, **High** — centered at 0° , 4° , and 6° , each with half-width 4° . The output variable is fan *Speed* (%), described by three triangular sets centered at 0% , 50% , and 100% , each with half-width 50% . Figure 2 plots both.

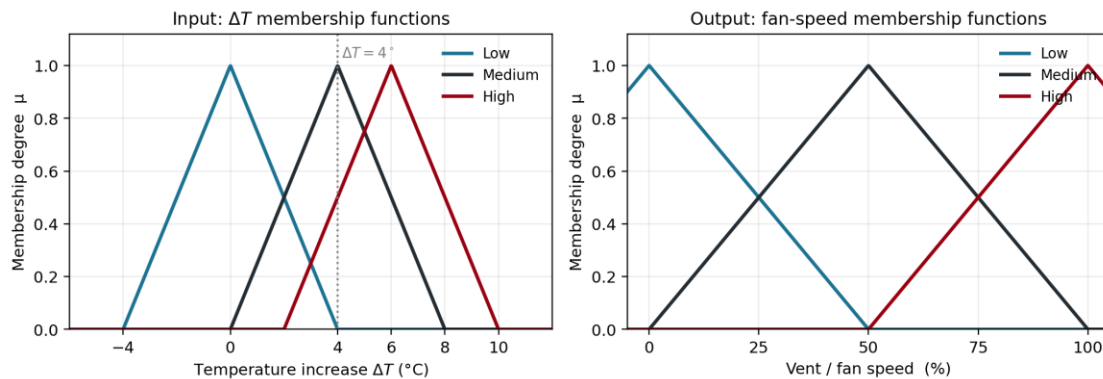


Figure 2. Input (ΔT) and output (*speed*) membership functions.

Every set above is built from the same **general triangular membership function**, with peak (center) c and half-width w :

$$\mu(x; c, w) = \max(0, 1 - |x - c| / w)$$

The Rule Base

We want rules of the form “*IF ΔT is ..., THEN Speed is ...*” — the fuzzy analog of an implication in classical logic, following the **Mamdani method** (Ebrahim Mamdani, 1975, one of the first practical fuzzy steam controllers):

Rule	IF ΔT is ...	THEN Speed is ...
R1	Low	Low
R2	Medium	Medium
R3	High	High

Defuzzification: the Centroid Formula

Once the output sets have been clipped by their rule's firing strength and combined (point-wise maximum) into one aggregated shape $\mu_{agg}(y)$, the crisp output is its **center of gravity**:

$$y^* = \int y \cdot \mu_{agg}(y) dy / \int \mu_{agg}(y) dy$$

In practice — and for this problem set — approximate the integral with a **discrete sum** over evenly spaced sample points y_i (e.g. every 10% of speed):

$$y^* \approx \sum_i [y_i \cdot \mu_{agg}(y_i)] / \sum_i \mu_{agg}(y_i)$$

How Clipping and Aggregation Work

Each rule's output set gets **clipped** — capped so its membership never exceeds that rule's firing strength, the fuzzy analog of *AND*. All the clipped sets are then **aggregated** by taking their point-wise maximum, the fuzzy analog of *OR*: at every point, the aggregated shape reflects whichever rule's evidence is stronger. The figure below works through both steps using firing strengths $w = 1.0$ and $w = 0.5$ — the values this worksheet's example actually produces.

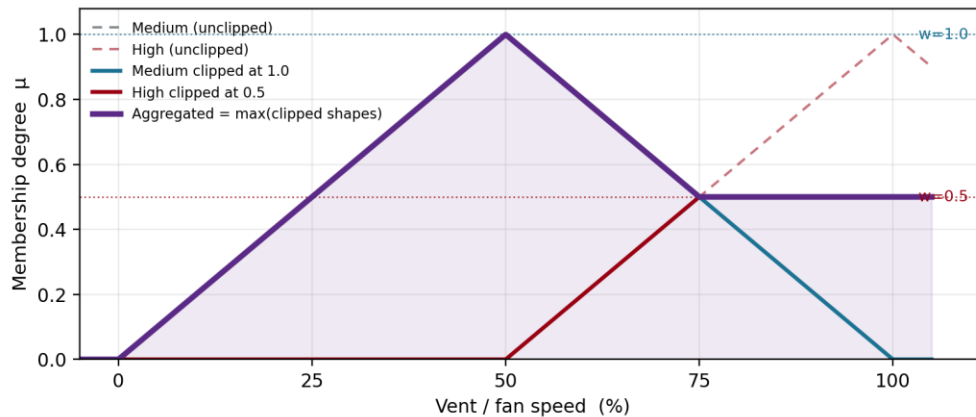


Figure 3. Clipping two output sets at strengths 1.0 and 0.5, then aggregating by taking the maximum at every point.

Two things to notice, since they generalize to any firing strengths: **(1)** a clipped set only goes flat where its *own* unclipped curve would have risen above its firing strength — elsewhere it's untouched. **(2)** wherever the two clipped curves cross, the aggregated (purple) curve switches from following one to following the other, since it always tracks whichever clipped value is *larger* at that point. The purple outline is the single fuzzy set you then defuzzify with the centroid formula above.

Problem 1. Write the membership functions

Using the general triangular formula above and Figure 2, write the piecewise formula for each of the three input sets, $\mu_{Low}(\Delta T)$, $\mu_{Medium}(\Delta T)$, and $\mu_{High}(\Delta T)$, stating the domain (range of ΔT) over which each piece applies.

Problem 2. Fuzzify the input

The room's temperature has increased by $\Delta T = 4^\circ\text{C}$. Using your formulas from Problem 1, compute $\mu_{Low}(4)$, $\mu_{Medium}(4)$, and $\mu_{High}(4)$. Which rule(s) from the rule base fire, and with what strength does each fire?

Problem 3. Defuzzify — calculate the resulting speed

Using your firing strengths from Problem 2, and following the same two-step procedure shown in Figure 3, apply the discrete centroid formula above. (Sample the aggregated shape at Speed = 0, 10, 20, ..., 100%, or integrate.)

Report the resulting fan speed as a percentage. Does the answer make intuitive sense given how strongly **Medium** and **High** fired?